

Mathematical and Physical Modeling for MATLAB Simulation of Entangled Photon Systems in Optical Fibers

Marco Cioci, 10694772

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1 General Framework

1.1 Purpose and Scope

The purpose of this document is to establish a structured working framework for the mathematical modeling and simulation of polarization-entangled photon pairs propagating in optical fibers.

This text is not intended to be part of the final thesis manuscript. Instead, it serves as an internal reference throughout the development process, supporting communication, ensuring consistency of notation, and enabling progressive validation of both theoretical and computational steps.

The primary objective is to organize the physical assumptions, mathematical tools, and simulation strategy in a coherent and reproducible manner, maintaining a direct correspondence between theoretical formulation and MATLAB implementation.

Simulation notes. At MATLAB simulator level, this document is treated as the reference specification for notation, model hierarchy, and implementation priorities. The simulator should therefore be organized so that each theoretical block corresponds to a clearly identifiable module, making it possible to validate intermediate results step by step and to extend the code without changing the whole structure.

1.2 Theoretical Formalism and Notation

The system is described within the formalism of quantum information theory. The relevant mathematical structures include Hilbert spaces, quantum states, tensor products, unitary operators, density operators, and quantum measurements. Classical optics is introduced only when necessary to motivate the physical origin of the effective quantum transformations.

The notation and formal tools are aligned, as far as possible, with the framework of *Quantum Computation and Quantum Information* by Nielsen and Chuang. In particular, the computational basis of a single qubit is identified with the polarization basis:

$$|0\rangle \equiv |H\rangle, \quad |1\rangle \equiv |V\rangle,$$

where $|H\rangle$ and $|V\rangle$ denote horizontal and vertical polarization states.

This choice fixes from the beginning the language in which the system is described: polarization states are treated as qubit states, bipartite photon pairs are treated as two-qubit systems, and optical transformations are represented by operators acting on finite-dimensional Hilbert spaces.

Simulation notes. The simulator should adopt from the beginning a quantum-information-oriented representation: state vectors for pure states, density matrices for mixed states, tensor-product composition for bipartite systems, and matrix operators for evolutions and measurements. The polarization encoding must be fixed once and for all as $|H\rangle \leftrightarrow |0\rangle$, $|V\rangle \leftrightarrow |1\rangle$, so that all MATLAB routines use the same basis convention.

1.3 Physical Scenario and Experimental Setting

The physical setting is as follows. A spontaneous parametric down-conversion (SPDC) source generates pairs of polarization-entangled photons, ideally prepared in a Bell state. The two photons are injected into distinct optical paths, denoted A and B , and propagate through independent optical fibers.

Each fiber acts on the polarization degree of freedom of the corresponding photon. In the most general description, this action is represented by a unitary operator on a single-qubit Hilbert space. However, as a first approximation motivated by experimental considerations, the dominant residual effect is modeled as a phase rotation around the computational Z -axis of the Bloch sphere.

The aim is therefore not to describe the full electromagnetic propagation problem from first principles at the initial stage, but rather to identify an effective quantum model that captures the physically relevant action of the fibers on polarization entanglement.

Simulation notes. At simulator level, the physical picture is reduced to the minimum set of objects needed for computation: an initial entangled two-qubit state, two labeled channels A and B , and one local transformation per channel. The first implementation should neglect unnecessary optical detail and describe each fiber only through its effective action on polarization.

2 Modeling Strategy

2.1 Progressive Modeling Approach

The modeling approach follows a progressive refinement strategy, in which the system is described through successive levels of approximation of increasing physical complexity.

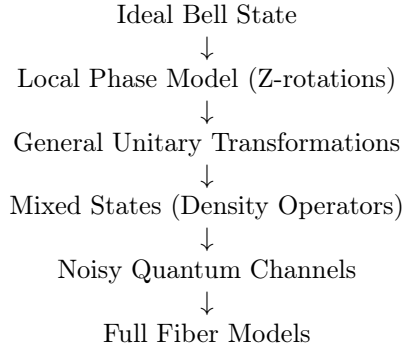
Each stage introduces additional effects while remaining consistent with the previous ones. This ensures both analytical transparency and modularity in the implementation.

From a theoretical perspective, this approach isolates the contribution of individual physical mechanisms. From a computational perspective, it enables a layered structure in the simulator, where each refinement corresponds to an additional module.

Simulation notes. The MATLAB code should be designed incrementally. A working minimal simulator must exist before more realistic effects are added. Each new level should extend the previous one rather than replace it, so that one can compare simplified and refined models under controlled conditions and verify where each physical effect begins to matter.

2.2 Hierarchy of Modeling Levels

The overall workflow can be summarized as:



More explicitly:

- **Level 1: Minimal model**
Pure, maximally entangled state. Fiber action reduced to a relative phase. Analytical dependence of correlations on a single parameter.
- **Level 2: General unitary dynamics**
Arbitrary polarization rotations described by local unitary operators.
- **Level 3: Mixed-state description**
Introduction of density operators to model decoherence and loss of purity.
- **Level 4: Noisy channels**
Effective quantum channels (e.g. depolarizing maps) to represent fiber-induced noise.
- **Level 5: Physical fiber models**
More detailed descriptions, such as concatenations of random unitary segments or stochastic models of polarization evolution.

This hierarchy provides both a conceptual roadmap and an implementation roadmap. The first levels are designed to remain analytically tractable and physically interpretable, while the later levels are intended to increase realism and to connect more directly with experimentally relevant imperfections.

Simulation notes. This hierarchy should be mirrored directly in the codebase. A convenient choice is to implement one MATLAB function or class layer for each level: state preparation, local unitary evolution, density-matrix conversion, channel application, and advanced fiber propagation. This makes later thesis work easier, because each refinement corresponds to a visible computational step.

2.3 Baseline Model: Pure-State Description

In the first stage, the global bipartite system is described as a pure state in $\mathbb{C}^2 \otimes \mathbb{C}^2$. Fiber propagation is modeled through local unitary transformations, with particular focus on relative phase shifts.

The main objective is to analyze how these phase shifts affect measurable two-photon correlations. The relevant observables are operators of the form

$$\sigma_i \otimes \sigma_j, \quad i, j \in \{x, y, z\},$$

as well as observables corresponding to arbitrary measurement directions on the Bloch sphere.

This baseline model is the natural entry point because it is simple enough to allow explicit formulas, direct visualization of the state dependence on propagation parameters, and immediate comparison between analytical and numerical results.

Simulation notes. The first operational simulator should work entirely with pure two-qubit states represented as 4×1 complex vectors. The initial tasks are: generate the input Bell state, apply local matrices U_A and U_B , and compute expectation values of selected observables. This level should already produce plots or tables showing how correlations vary with the relative phase.

2.4 Extension to Mixed-State Formalism

At a more advanced stage, the model incorporates decoherence and depolarization effects. In this regime, the pure-state description becomes insufficient, and density operators are required.

This framework allows one to describe:

- reduced states of individual photons,
- loss of coherence due to frequency-dependent propagation,
- noisy quantum channels acting on polarization.

It also enables the computation of quantities such as fidelity, concurrence, and other entanglement measures.

The introduction of mixed states is therefore not a merely formal refinement. It is required whenever one wishes to describe partial information, environmental coupling, statistical mixtures, or effective non-unitary dynamics arising from unresolved physical degrees of freedom.

Simulation notes. Once the pure-state stage is stable, the simulator should be extended so that every relevant routine can also accept density matrices. In practice, this means introducing 4×4 matrix representations, partial trace routines, noise-channel application, and metric-evaluation functions. The pure-state code should remain available, but it should become a special case of the density-operator framework whenever possible.

2.5 Simulator Design and Architecture

The simulator is designed as a modular implementation of the theoretical model.

The initial modules will:

- generate ideal Bell states,
- apply local fiber transformations,
- compute correlation functions.

Subsequent modules will:

- introduce noise channels,
- compute reduced density operators,
- evaluate observables and entanglement measures,
- extract experimentally relevant quantities such as Bell parameters and Stokes-like expectation values.

The theoretical development is therefore directly coupled to the computational structure, ensuring consistency, flexibility, and traceability of results.

The architecture should distinguish clearly between reusable physical kernels and higher-level experiment scripts. In this way the simulator can evolve from a set of exploratory routines into a compact and reusable research tool.

Simulation notes. The architecture should be kept explicitly modular: one block for state preparation, one for channel action, one for observables, one for diagnostics, and one for post-processing. At MATLAB level, this suggests separating reusable numerical kernels from scripts used only for running specific experiments, so that the simulator can later become a small reusable library.

3 Physical and Mathematical Representation

3.1 Quantum System and Hilbert Space Structure

The starting point of the model is a source of polarization-entangled photon pairs generated by spontaneous parametric down-conversion (SPDC), in a configuration in which the two output paths are denoted A and B . When the relevant optical alternatives are made indistinguishable, the source can ideally prepare one of the Bell states of two polarization qubits.

In the present work, the natural encoding of the qubit is the polarization degree of freedom:

$$|H\rangle \equiv |0\rangle, \quad |V\rangle \equiv |1\rangle.$$

Accordingly, the computational basis of the bipartite system is

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

that is,

$$\{|H\rangle_A |H\rangle_B, |H\rangle_A |V\rangle_B, |V\rangle_A |H\rangle_B, |V\rangle_A |V\rangle_B\}.$$

The Hilbert space of the total system is therefore

$$\mathcal{H}_{AB} = \mathcal{H}_A \otimes \mathcal{H}_B \cong \mathbb{C}^2 \otimes \mathbb{C}^2,$$

which has dimension 4. As a consequence, a pure state of the two-photon system is represented by a complex vector with four components in the computational basis, while a general linear operator acting on the full system is represented by a 4×4 matrix.

This representation fixes the elementary mathematical objects used throughout the model: single-photon polarization states are qubit vectors in $\mathbb{C}^2$, two-photon states live in the tensor-product space $\mathbb{C}^2 \otimes \mathbb{C}^2$, and observables or evolutions are represented by matrices acting on these spaces.

Simulation notes. At MATLAB level, the natural internal representation is therefore immediate: a bipartite pure state is a complex vector of length 4, and bipartite operators are 4×4 matrices. The simulator should store basis ordering explicitly as

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

since all subsequent tensor products, observables, and density matrices depend on this convention.

3.2 Bell-State Initialization

As a reference initial condition, we consider the Bell state

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|H\rangle_A |V\rangle_B + |V\rangle_A |H\rangle_B),$$

which, in computational notation, is written as

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} (|01\rangle + |10\rangle).$$

In the ordered computational basis

$$\{|00\rangle, |01\rangle, |10\rangle, |11\rangle\},$$

this state is represented by the column vector

$$|\Psi^+\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \\ 0 \end{pmatrix}.$$

This state is maximally entangled and constitutes the natural input of the first version of the simulator. It is also the reference state with respect to which quantities such as fidelity and Bell-inequality violation will later be evaluated.

The choice of a Bell state is natural because it provides both a clean theoretical benchmark and a physically relevant entangled resource. It also allows the effect of the fibers to be interpreted directly in terms of changes in entanglement, coherence, and two-photon correlations.

Simulation notes. The simulator should include a dedicated state-preparation routine returning the Bell vector in canonical basis ordering. This reference state should be used systematically for debugging, verification of normalization, comparison with analytical formulas, and later evaluation of fidelity after propagation through the fibers.

3.3 Fiber Propagation as Local Unitary Evolution

Each photon propagates through its own optical fiber. At the level of the polarization qubit, the action of each fiber is described by a linear operator acting locally on the corresponding single-photon Hilbert space. In the most general ideal lossless description, such an operator is unitary. Therefore, if U_A and U_B denote the single-qubit operators associated with the two fibers, the evolution of the initial two-photon state is

$$|\psi_{\text{out}}\rangle = (U_A \otimes U_B) |\psi_{\text{in}}\rangle.$$

This expression is the natural bipartite extension of single-qubit evolution and follows directly from the tensor-product structure of composite quantum systems. It provides the correct starting point for any subsequent refinement of the model, including random unitary perturbations, concatenated fiber sections, or effective channel descriptions.

In this formulation, the two fibers act independently on the two polarization subsystems, and the total action is obtained by tensor composition of the corresponding local transformations. This is the mathematically appropriate description whenever the fibers do not generate direct interaction between the two photons.

Simulation notes. The fundamental evolution primitive in MATLAB should be the tensor-product action

$$\psi_{\text{out}} = (U_A \otimes U_B) \psi_{\text{in}}.$$

It is convenient to isolate this in a single function, so that later changes in the single-fiber model automatically propagate to the full bipartite simulation.

3.4 Reduced Phase Model

Although the most general fiber action may involve arbitrary polarization rotations, birefringence, and mode-coupling effects, the first approximation adopted here is simpler and is directly motivated by the experimental control scheme. After partial compensation of polarization drifts, the dominant residual effect is modeled as a relative phase shift between the horizontal and vertical components, that is, as an effective rotation around the Z -axis of the Bloch sphere.

At the single-qubit level, this is represented by a diagonal unitary of the form

$$U(\theta) = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix},$$

up to an irrelevant global phase. If the two fibers induce phases θ_A and θ_B , then the total transformation is

$$U_A \otimes U_B = U(\theta_A) \otimes U(\theta_B).$$

Applying this operator to the reference Bell state $|\Psi^+\rangle$ gives

$$|\psi(\theta_A, \theta_B)\rangle = \frac{1}{\sqrt{2}} (e^{i\theta_B} |01\rangle + e^{i\theta_A} |10\rangle).$$

By factoring out a global phase, which has no physical effect on measurement statistics, this state can be rewritten as

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} (|01\rangle + e^{i\theta} |10\rangle),$$

where

$$\theta = \theta_A - \theta_B.$$

Thus, in the first effective model, the action of the two fibers is reduced to a single physically relevant parameter: the relative phase θ between the two components of the entangled state. This parameter is expected to depend on propagation conditions such as birefringence, path length,

and environmental fluctuations, and it will be the main control variable in the first stage of the simulator.

This reduced description is especially useful because it isolates the effect of interest while preserving the essential entangled two-qubit structure. It is therefore a controlled approximation rather than an arbitrary simplification.

Simulation notes. This is the first concrete propagation model to implement. The simulator should expose θ_A and θ_B as input parameters, but internally recognize that the observable behavior depends only on $\theta = \theta_A - \theta_B$. In practice, it is useful to keep both forms available: the two-phase description for physical interpretation, and the reduced single-parameter description for plots, parameter sweeps, and analytical checks.

3.5 Interpretation and Role of the Reduced Model

The reduced phase model is particularly convenient for three reasons. First, it preserves the pure-state description, allowing the initial implementation to remain simple and analytically transparent. Second, it isolates the effect that is most directly relevant for the first experimental analysis, namely the deformation of two-photon correlations induced by a relative phase. Third, it prepares the transition to more general models, since the same bipartite formalism can later be extended to density operators and noisy channels without changing the conceptual structure of the problem.

For this reason, the first stage of the simulator will be built around the family of states

$$|\psi(\theta)\rangle = \frac{1}{\sqrt{2}} (|01\rangle + e^{i\theta}|10\rangle),$$

from which one will compute expectation values of local and nonlocal observables, correlation functions, and Bell-type quantities as functions of θ .

The role of this model is therefore twofold. On the one hand, it is a physically motivated approximation to the propagation problem. On the other hand, it is the minimal fully operational model from which one can derive and test measurable predictions in a clean and controlled way.

Simulation notes. The first validated MATLAB study should therefore be a parameter sweep in θ , with numerical evaluation of state vectors, expectation values, and correlation quantities. This gives a minimal but complete simulator: one input family of states, one effective propagation parameter, and a set of measurable outputs that can be directly plotted and compared against theory.

4 References

- M. A. Nielsen and I. L. Chuang, *Quantum Computation and Quantum Information*, Cambridge University Press.